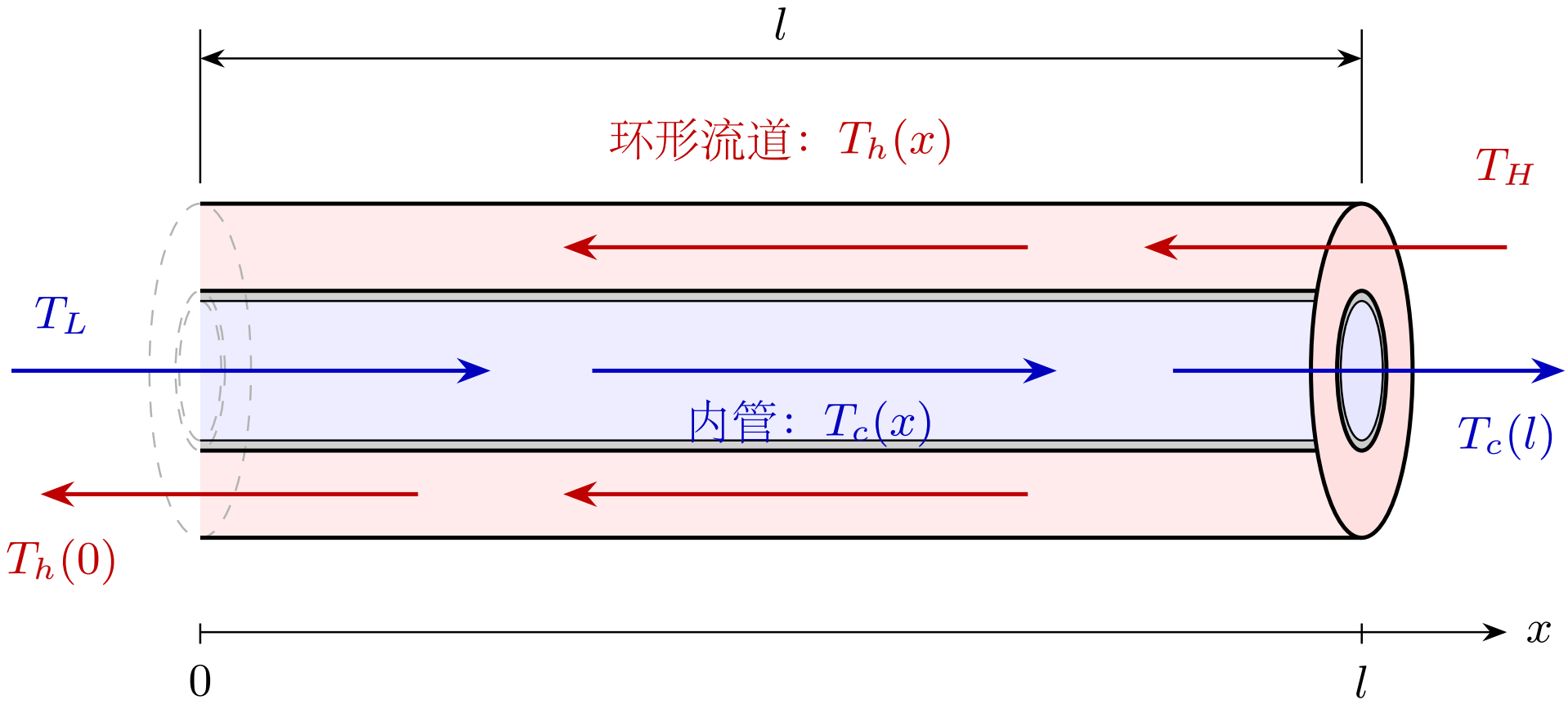
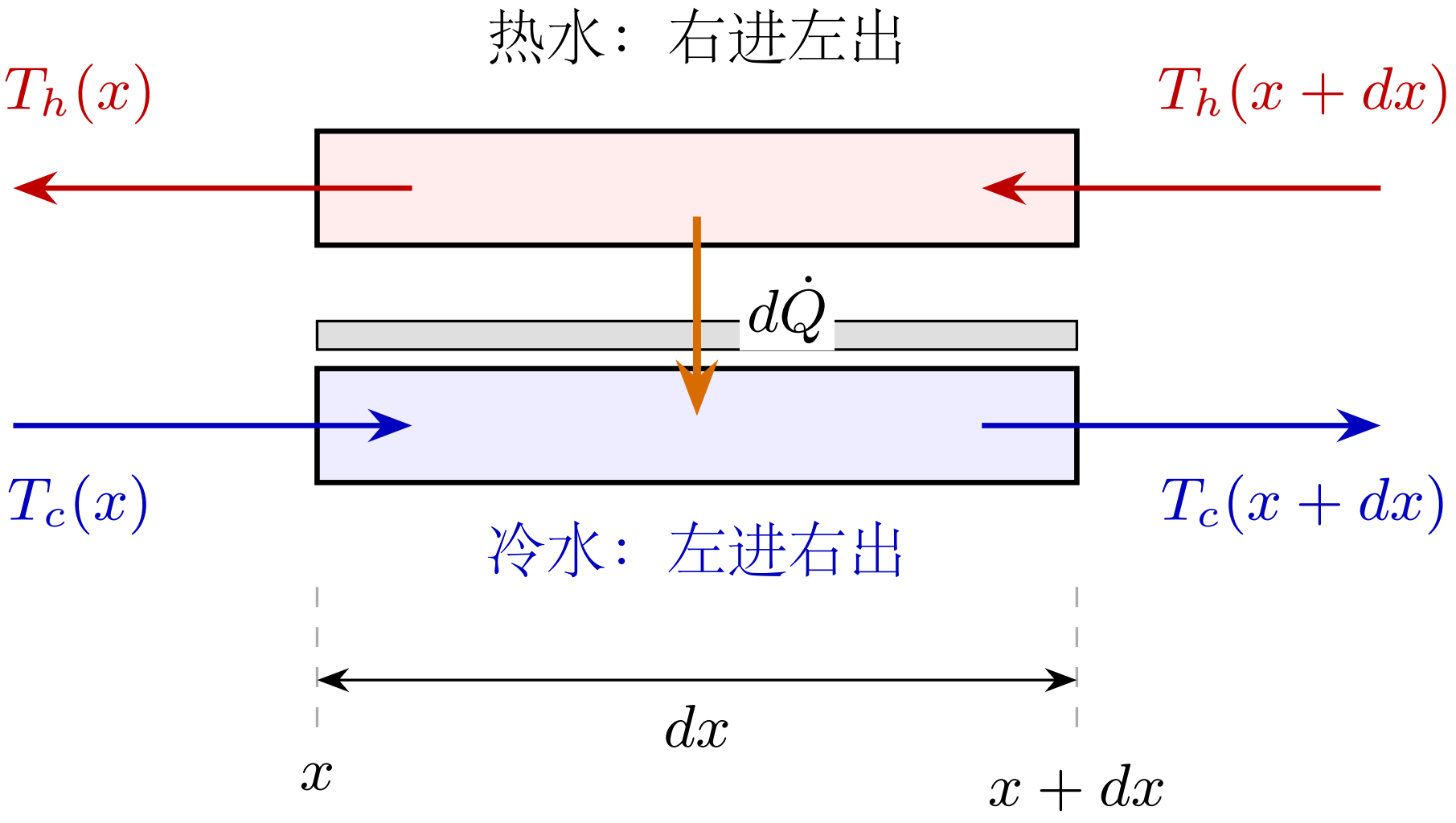
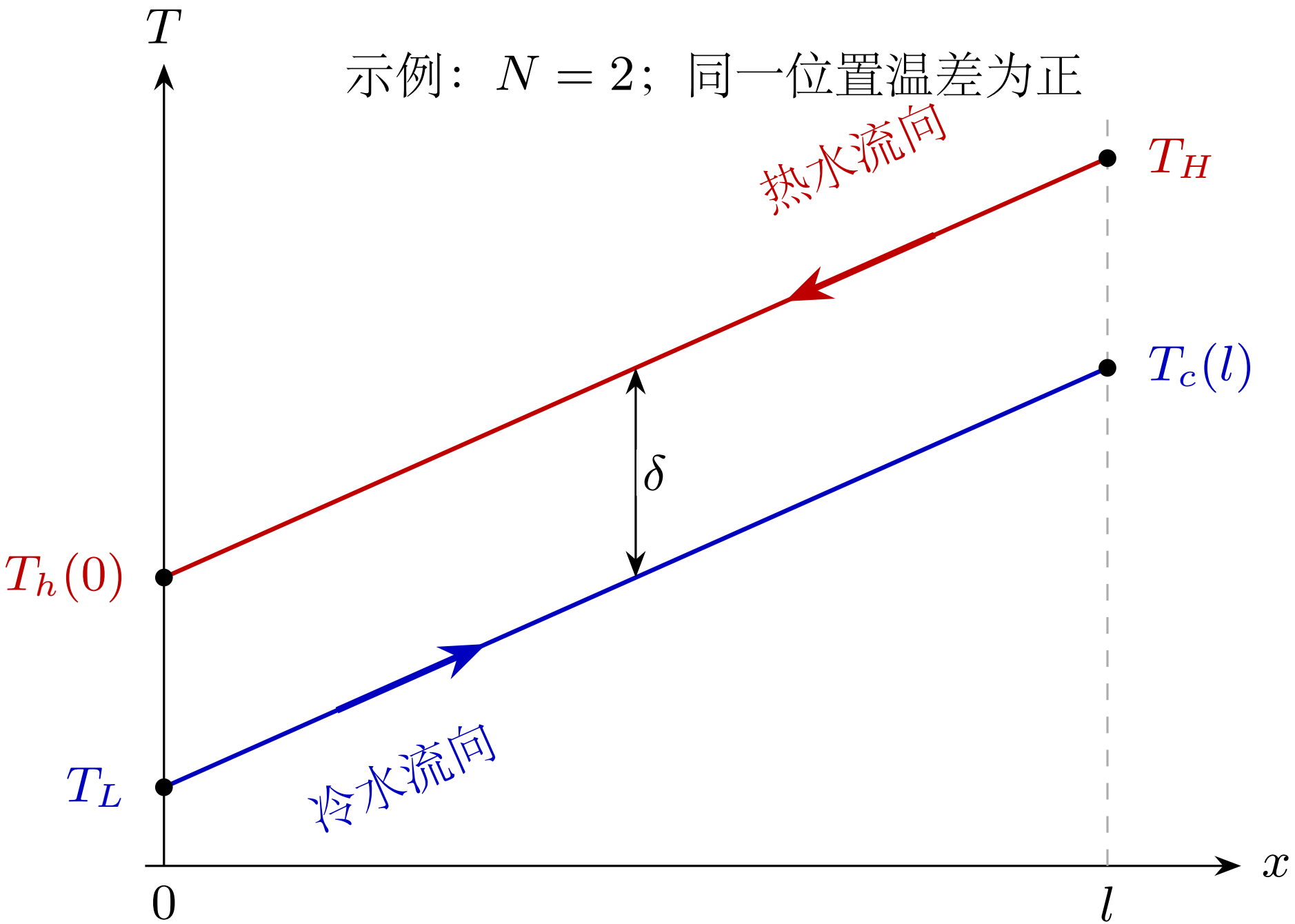
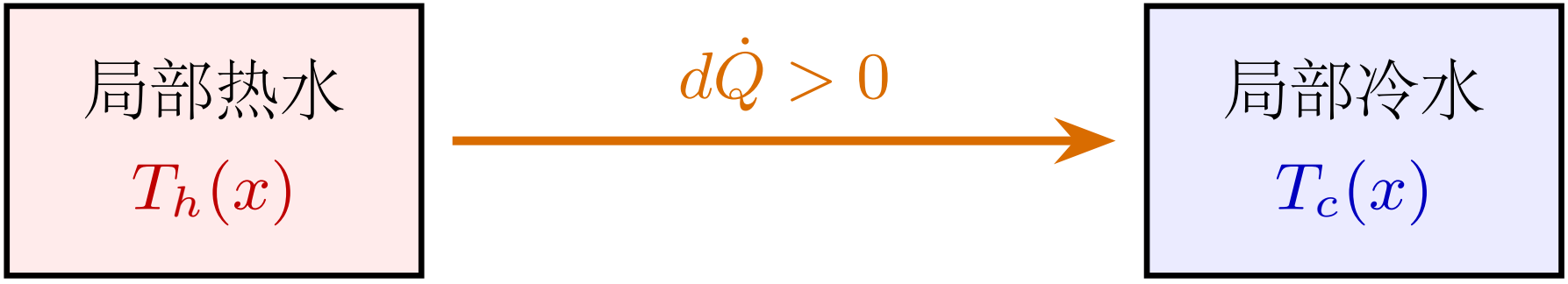


外管透明示意；冷热水由中间管壁分隔

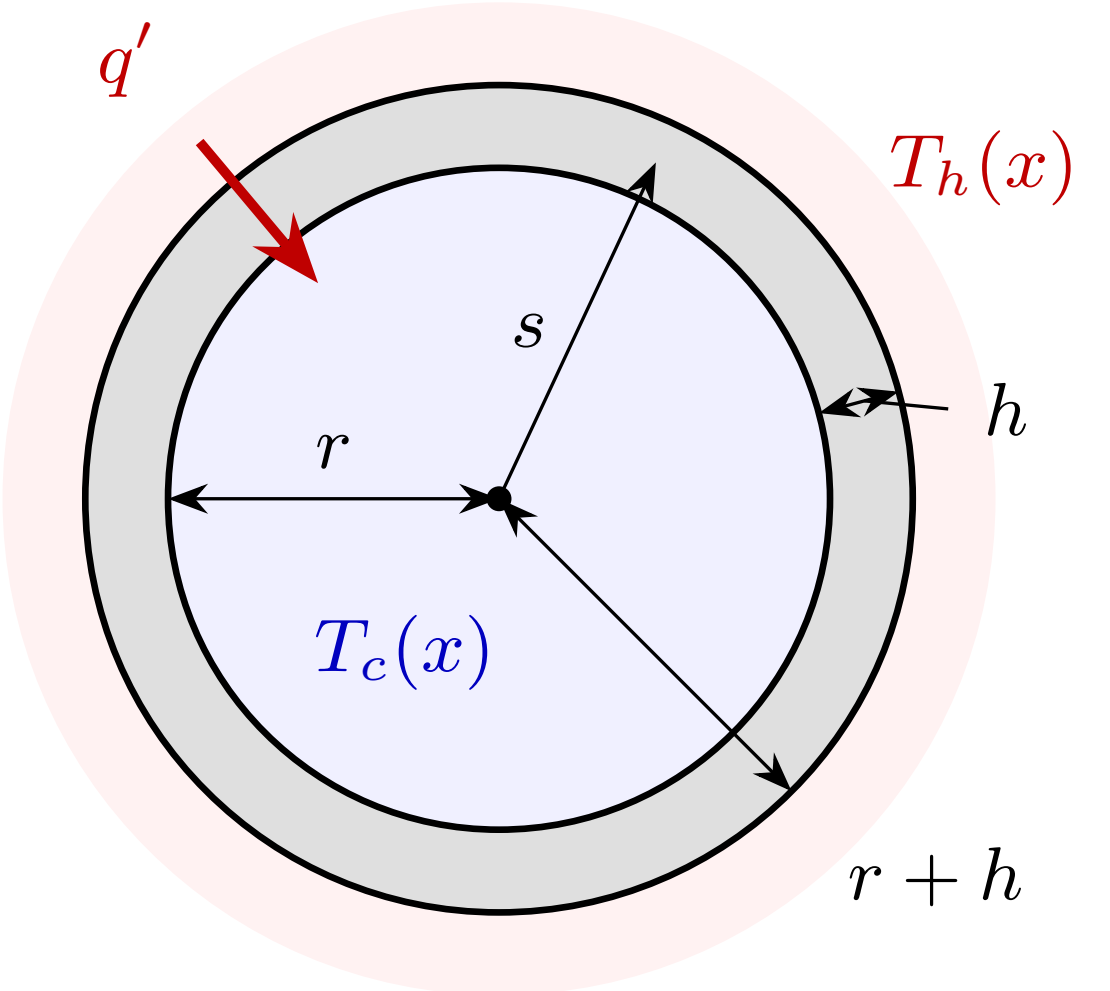

热水：右进左出
$T_h(x)$
$T_h(x+dx)$
$d\dot{Q}$
$T_c(x)$
$T_c(x+dx)$
冷水：左进右出
dx
x
$x+dx$


T
示例：$N = 2$；同一位置温差为正
热水流向
T_H
$T_c(l)$
δ
$T_h(0)$
T_L
冷水流向
0
l
x



$$d\dot{S}_h = -d\dot{Q}/T_h \qquad d\dot{S}_c = d\dot{Q}/T_c$$

$$d\dot{S}_{\text{gen}} = d\dot{Q}(1/T_c - 1/T_h) > 0$$



径向截面；壁厚示意放大